Encoding Glue Functions with TBNs

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Background

A *thermodynamic binding network* (TBN) models molecular building blocks (*monomers*) carrying labeled binding sites (*domains*). It has no geometry, and each domain a binds only its exact complement a^*. Groups of monomers form *polymers*. The score $E(C) = -w\,b(C) - p(C)$ rewards bonds $b(C)$ and separate polymers $p(C)$; lower energy is more favorable [1].

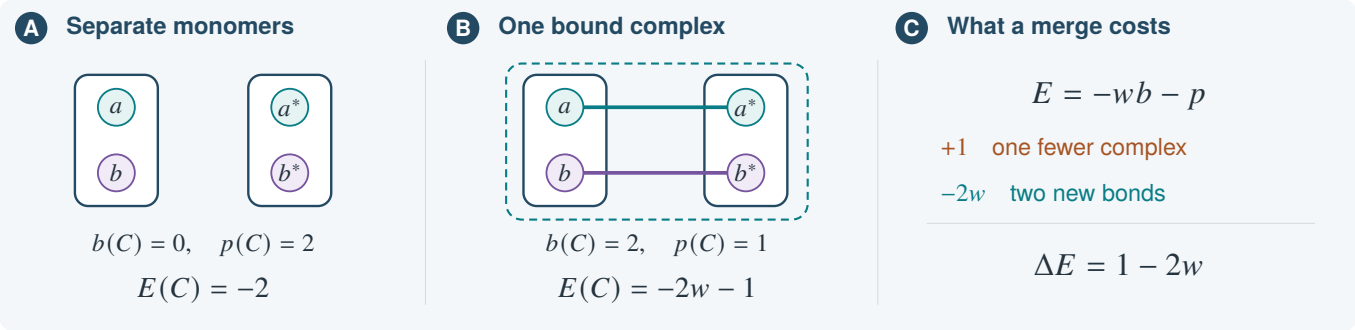


Figure 1. Rounded boxes are monomers, labels are domains, and lines are complementary bonds. The dashed boundary marks one polymer. Positions are illustrative: TBNs have no geometry.

TBNs can indirectly realize any affinity function allowed by *thermodynamic affinity networks* (TANs), using auxiliary monomers to reproduce the desired stable configurations [2]. Can specific glue behaviors instead have simple, *direct* implementations, without the general whole-system encoding?

The problem

Using only ordinary complementary domains underneath, construct implementations of:

1. **Larger binding strengths.** Encode contacts with different strengths. How does the binding benefit scale with the number of domains used?
2. **Flexible, non-complementary binding.** Let one logical site bind to several possible partners—for example, a to either b or c—with specified strengths. Preserve its intended capacity: alternatives must not become extra sites.
3. **Negative, repulsive binding.** Make a designated interaction impose an energy penalty that offsets favorable contacts. Absence of a complementary bond alone is not a negative strength.

Seek helper-free implementations first; explain any helpers that are needed. Define what represents a logical glue and how its strength is measured. Does the behavior survive when multiple copies or several glue types interact?

A starting point

Use Figure 1 to explain **cooperative binding**: a merge forming m new bonds changes the energy by $1 - mw$. For $\frac{1}{2} < w < 1$, two bonds make a merge favorable, but one does not. Then tackle one behavior above. Aim for a small gadget, its energy table, and a readable correctness argument—or an impossibility proof under explicit restrictions. AI-assisted search can find candidates; explain *why* they work.

References

[1] K. Breik, C. Chalk, D. Doty, D. Haley, and D. Soloveichik. Programming Substrate-Independent Kinetic Barriers with Thermodynamic Binding Networks. *IEEE/ACM Trans. Comput. Biol. Bioinform.* 18(1):283–295, 2021.

[2] A. K. A. Yilmaz, D. Soloveichik, and A. Luchsinger. Thermodynamic Binding Networks Capture Equilibrium Monomer-Polymer Thermodynamics. *DNA 32*, LIPIcs 387, 9:1–9:20, 2026.